PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

(All Branches NBA Accredited)

Department of Information Technology

Academic Year: 2025-26
Semester: IV
Class / Branch: SE- IT A
Subject:Network design lab
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Date of Submission:08/04/2026

EXPERIMENT NO 10

AIM: To design and implement connection less network application through Socket API.

OUTPUT:

1) Client start

```
nagios@worker01:/home/kbd/socket$ cc -o udpc udpc.c
nagios@worker01:/home/kbd/socket$ ./udpc
Enter a number: 5
Factorial = 120
nagios@worker01:/home/kbd/socket$
```

2) Client start

```
nagios@worker01:/home/kbd/socket$ ./udps
UDP Server is ready...
Received number: 5
Result sent: 120
```

3) PACKETS CAPTURED (sudo tcpdump -l lo -n -xX port 8000)

```
[sudo] password for nagios:
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on lo, link-type EN10MB (Ethernet), capture size 262144 bytes
09:32:47.363535 IP 192.168.104.26.48880 > 192.168.104.26.8000: UDP, length 4
        0x0000:  4500 0020 8b13 4000 4011 5e34 c0a8 681a  E.....@.@.^4..h.
        0x0010:  c0a8 681a bef0 1f40 000c 51a3 0500 0000  ..h....@..Q.....
09:32:47.363695 IP 192.168.104.26.8000 > 192.168.104.26.48880: UDP, length 8
        0x0000:  4500 0024 8b14 4000 4011 5e2f c0a8 681a  E..$..@.@.^/..h.
        0x0010:  c0a8 681a 1f40 bef0 0010 51a7 7800 0000  ..h..@....Q.x...
        0x0020:  0000 0000                                ....
```

4) NETSTAT

```
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
udp        0      0 0.0.0.0:8000            0.0.0.0:*
```

Conclusion

The implementation of a connectionless network application using the socket API shows how data can be transmitted quickly and efficiently without the need to establish a prior connection. It emphasizes the use of protocols such as UDP, where speed and low latency are given more importance than guaranteed delivery and ordered transmission of data.